ME 5309 –

Mid-Semester Examination

Duration: 1 hour **Maximum marks:** 30 **Open notes**

Roll No.: _____ Name: _____

Return the question paper with the answer sheet after filling details.

Instructions

1. **Answer any THREE of the four questions.** Each question carries **10 marks**.
2. If you attempt more than three, all attempts will be read and the **best three** will be counted. Clearly strike out any work you do not want considered.
3. This is an **open-notes** examination. Your own handwritten notes only are permitted. Textbooks, laptops, phones, internet access and the exchange of any material between candidates are **not** permitted.
4. Non-programmable calculators are allowed.
5. **Show the formula you are using and the substitution into it.** A final numerical answer without the formula being written earns no credit, even if correct. Wherever a formula is taken from your notes, you need not derive it unless the question asks for a derivation.
6. State any assumption you make, and give the units of every numerical answer.
7. Give numerical answers to at least three significant figures. Take $g = 9.81\ \mathrm{m/s^2}$.
8. Each question is designed to take about 20 minutes with open notes and a calculator.
9. **As announced, malpractice can result in disciplinary actions as per the Senate mandate.**

Question 1 [10 marks]

Answer all three parts within this Ques 1. Each part is independent of the others.

(a) A machine-vision camera inspects parts moving on a conveyor. The sensor is 1280 pixels wide with square pixels of 5 μm, the lens has a focal length of 16 mm, and the camera looks down on the belt from a standoff of 400 mm.

(i) Determine the focal length f_x **in pixels**.

(ii) Determine the width of the field of view on the belt, and hence the distance on the belt that corresponds to one pixel. **[4]**

(b) A force sensor of sensitivity 10 mV/N is read by a 12-bit ADC whose input range is 0 V to 5 V. Determine the smallest change in force that the system can resolve. **[3]**

(c) The first-order plant

$$\dot{x} = -2x + u$$

is to be implemented in a digital controller. Using a zero-order hold on the input and a sampling period $T = 0.1$ s, obtain the **exact** discrete-time model

$$x_{k+1} = F\,x_k + G\,u_k$$

and give F and G numerically. State the formulae you use for F and G before substituting. **[3]**

Question 2 [10 marks]

A single revolute joint moving in the **vertical** plane is described by

$$J\ddot{\theta} + b\,\dot{\theta} + mg\ell\sin\theta = \tau,$$

where θ is measured from the **downward** vertical and

$$J = 0.02\ \text{kg·m}^2, \qquad b = 0.05\ \text{N·m·s/rad}, \qquad m = 1\ \text{kg}, \qquad \ell = 0.25\ \text{m}.$$

(a) Choose a state vector and write the nonlinear state-space model $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \tau)$. For a **constant** applied torque τ_0, write the equation that determines the equilibria and state the range of τ_0 for which an equilibrium exists. **[2]**

(b) Linearise the model about the equilibrium at $\theta_e = 60^\circ$ and give the matrices A and B numerically. **[2]**

(c) The joint is now controlled by pure PD feedback,

$$\tau = K_p\,(\theta_d - \theta) - K_d\,\dot{\theta}, \qquad K_p = 50\ \text{N·m/rad},$$

with a constant commanded (desired) angle $\theta_d = 90^\circ$. Show that the joint does **not** come to rest at θ_d. Write the equation satisfied by the steady-state angle θ_{ss}, and hence estimate the steady-state error in degrees. **[4]**

(d) A gravity-compensation term is now added, so that

$$\tau = K_p\,(\theta_d - \theta) - K_d\,\dot{\theta} + mg\ell\sin\theta.$$

State the new steady-state error, and state with justification whether this additional term changes the closed-loop poles of the linearisation about θ_d. **[2]**

Question 3 [10 marks]

A joint is instrumented with a position encoder only, and its model is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

with the state $\mathbf{x} = \begin{bmatrix} \theta & \dot{\theta} \end{bmatrix}^{\mathsf{T}}$. The control loop runs at $f_s = 1$ kHz. A velocity signal is required for a PD control law.

(a) State the two distinct reasons why forming velocity by raw finite differencing of successive encoder readings gives a poor estimate. **[2]**

(b) A first-order low-pass filter with cut-off frequency $f_c = 20$ Hz is to be applied to the differenced signal. Discretise the filter using the bilinear (Tustin) transform at $f_s = 1$ kHz and write the resulting difference equation with numerical coefficients. Verify that its DC gain (steady state ratio between desired command and actual output) is unity. **[2]**

(c) Instead of filtering, a Luenberger (full state observer covered in class) observer

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + Bu + L\,(y - C\hat{\mathbf{x}})$$

is to be used. **Derive** the characteristic polynomial of $A - LC$ in terms of $L = \begin{bmatrix} \ell_1 & \ell_2 \end{bmatrix}^{\mathsf{T}}$, and hence determine L so that both estimator poles lie at $s = -10$. **[4]**

(d) Both the filter of part (b) and the observer of part (c) produce a velocity estimate. State one advantage the observer has that no filter of this kind can provide, and state one circumstance in which the filter would be the safer choice. **[2]**

Question 4 [10 marks]

A linearised, gravity-loaded joint, with the input gains lumped into a single constant b, is described by

$$\ddot{\theta} + a\,\dot{\theta} + c\,\theta = b\,u, \qquad a = 2, \quad c = 6, \quad b = 1,$$

where c arises from the gravity load. The nominal controller is state feedback with reference feedforward,

$$u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}.$$

(a) With $k_1 = k_2 = 2$, find the closed-loop poles and determine the value of N_{ff} for which the DC gain from r to θ is exactly unity. **[2]**

(b) The actual actuator gain turns out to be **25% lower** than assumed, that is $b = 0.75$, while N_{ff}, k_1 and k_2 are left at the values found in part (a). Compute the steady-state value of θ for a unit step reference $r = 1$ and the resulting percentage tracking error. **[2]**

(c) The plant is now augmented with the integral of the tracking error,

$$\dot{\xi} = r - \theta, \qquad u = -k_1\theta - k_2\dot{\theta} + K_I\,\xi.$$

Derive the closed-loop characteristic polynomial **symbolically** in terms of a, b, c, k_1, k_2 and K_I. Hence place the closed-loop poles at $s = -4$ and $s = -2 \pm 2j$, and determine k_1, k_2 and K_I. **[4]**

(d) Show that the controller designed in part (c) gives **zero** steady-state error for **any** values of b and c, provided the closed loop remains stable. Name one practical failure mode of integral action and the standard remedy for it. **[2]**

— End of question paper —